

SURAJ BHARDWAJ

AI Researcher & Data Scientist

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[LinkedIn](#) | [GitHub](#)

Summary

AI and data science professional with expertise in deep learning, LLMs, and computer vision. Experienced in developing scalable, research-driven ML systems and deploying them using modern MLOps tools.

Skills

Programming, Advanced

[Python](#) [C++](#) [SQL](#) [MATLAB](#) [Bash](#) [Git](#) [Linux](#)

Machine Learning & Deep Learning, Expert

[Self-Supervised Learning](#) [Large Language Models \(LLMs\)](#) [Transformers](#) [Graph Neural Networks](#) [Diffusion Models](#)
[Variational Autoencoders](#) [SparseCNNs](#) [LSTMs](#) [Object Detection](#) [Image Segmentation](#)

Frameworks & Libraries, Proficient

[PyTorch](#) [TensorFlow](#) [Keras](#) [HuggingFace](#) [OpenCV](#) [Ray](#) [FAISS](#) [LangChain](#) [ChromaDB](#) [Timm](#) [Optuna](#) [Scikit-learn](#)

DevOps & Deployment, Intermediate

[Docker](#) [AWS](#) [GitHub Actions](#) [GitLab CI/CD](#) [MLflow](#) [FastAPI](#) [Flask](#)

Data Visualization, Proficient

[Matplotlib](#) [Seaborn](#) [Plotly](#) [Pandas](#) [NumPy](#) [Tableau](#) [Power BI](#) [Open3D](#) [Trimesh](#)

NLP & LLMs, Advanced

[LangChain](#) [Ollama](#) [Transformers](#) [SpaCy](#) [NLTK](#) [RAPTOR](#) [SentenceTransformers](#)

Experience

Research Assistant (KARLI and SALSA Projects) @ Fraunhofer IOSB (2023-05-01 - 2025-01-01)

<https://www.iosb.fraunhofer.de/>

Conducted AI research for human-machine interaction and autonomous driving projects (KARLI, SALSA).

- Developed an LLM-based chatbot using RAG and RAPTOR for scalable retrieval in knowledge-intensive settings.
- Designed the Clustered Feature Weighting (CFW) algorithm to mitigate class imbalance in multimodal datasets.
- Contributed to VoxelNeXt deployment for LiDAR-based 3D object detection on the Pandaset dataset.
- Led technical demos of a Level 3 Mercedes-Benz Occupant Monitoring System for stakeholders.
- Worked with Python, PyTorch, Ray, FAISS, RAPTOR, LangChain, Docker, Streamlit, and AWS.

Master Thesis Researcher – Improved Driver Distraction Detection using Self-Supervised Learning @ Fraunhofer IOSB (2023-11-01 - 2024-05-01)

<https://www.iosb.fraunhofer.de/>

Investigated cross-modality generalization using self-supervised vision transformers for driver distraction detection.

- Innovated the Clustered Feature Weighting (CFW) algorithm using HDBSCAN for long-tailed distribution.
- Achieved 7.17% performance improvement over supervised baselines using DINOv2-pretrained ViT-B/14.
- Used tools like Ray, PyTorch, Scikit-learn, OpenCV, and Git for reproducible experimentation.

Education

Department of Electrical Engineering and Computer Science, M.Sc. Mechatronics (2019-10-01 - 2025-30-01)

[University of Siegen](#), score: GPA: 2.1 (German Scale)

Courses: Software Engineering, Unsupervised Deep Learning, Artificial Intelligence, Mechatronic Systems

Department of Mechanical Engineering, B.Tech. Mechanical Engineering (2014-08-01 - 2018-05-31)

[National Institute of Technology Hamirpur](#), score: GPA: 2.5 (German Scale)

Courses: Engineering Mathematics, Numerical Methods, Data Structures

Awards

NSTEDB Research Grant (2018-11-01)

National Science & Technology Entrepreneurship Development Board (NSTEDB), India

Awarded for the research project 'Effect of Size and Cascading of Receivers on the Performance of a Solar Collector System'. Recognized for innovation in solar energy technology.

SJVN Silver Jubilee Merit Scholarship (2014-06-01)

SJVN Foundation

Granted for securing 21st rank in Himachal Pradesh Board examinations.

Certificates

Modern Computer Vision™ PyTorch, Tensorflow2, Keras & OpenCV4 (2023-03-20)

Udemy (Instructor: Rajeev D. Ratan)

Unsupervised Learning, Recommenders, Reinforcement Learning (2023-03-18)

DeepLearning.AI & Stanford University (Coursera)

Advanced Learning Algorithms (2022-08-27)

DeepLearning.AI & Stanford University (Coursera)

Supervised Machine Learning: Regression and Classification (2022-07-21)

DeepLearning.AI & Stanford University (Coursera)

Machine Learning (2020-08-12)

Stanford University (Coursera, Instructor: Andrew Ng)

Volunteers

Conference Reviewer @ IEEE ITSC 2025 (International Conference on Intelligent Transportation Systems) (2025-03-01 - 2025-07-01)

<https://ieee-itsc.org/2025/>

Served as a peer reviewer for the IEEE ITSC 2025 conference in the domains of machine learning and computer vision.

- *Reviewed submissions in unsupervised learning, segmentation, and safe motion generation.*
 - *Focused on evaluating methods in ML, CV, and intelligent systems for transportation and autonomous driving.*
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Publications

Self-supervised Driver Distraction Detection for Imbalanced Datasets @ IEEE 28th International Conference on Intelligent Transportation Systems (ITSC 2025) (2025-11-01)

<https://ieee-itsc.org/2025/>

Proposed a novel unsupervised adaptation strategy to address class imbalance and enhance generalization in computer vision tasks — directly applicable to clinical AI settings with limited or unlabelled data.

Effect of Size and Cascading of Receivers on the Performance of a Solar Collector System @ Springer, Singapore (2020-02-01)

https://doi.org/10.1007/978-981-15-0124-1_113

This study examines how cascading configurations in parabolic dish solar collectors improve collection efficiency by 3.2%–5.4%, highlighting the impact of receiver size and layout on thermal performance.

Projects

YOLO-Based Real-Time Object Detection in CARLA (2024-02-01 - 2024-09-01)https://github.com/SurajBhar/studienarbeit_repository

- Simulated a sensor suite with 8 cameras, radar, and 12 ultrasonic sensors at 30 FPS.
- Trained and evaluated 12 YOLO models on a custom dataset; YOLO8-m and YOLO11-m achieved best mAP@₍₅₀₋₉₅₎ scores.
- Integrated the fine-tuned YOLO11-m model into the CARLA pipeline for live inference on cars, pedestrians, and traffic signs.

RAG Chatbot using NVIDIA NIM (2024-03-01 - 2024-05-01)https://github.com/SurajBhar/rag_nim

- Utilized RAPTOR, FAISS, and ChromaDB for retrieval-augmented generation.
- Designed for scalable, domain-specific document understanding.

Codeninja: AI Code Assistant (2024-01-01 - 2024-03-01)<https://github.com/SurajBhar/codeninja>

- Integrated with a custom model served locally.
- Provided contextual, real-time code generation and fixes.

Implicit 3D Model Generation (2022-12-01 - 2023-04-01)<https://github.com/SurajBhar/ndf>

- Trained on ShapeNet Cars using SLURM on OMNI cluster.
- Implemented vector quantization and feature bottlenecks.

OOD Robustness with AugMix (2022-10-01 - 2022-11-01)<https://github.com/SurajBhar/augmix>

- Applied transfer learning and tuning with cosine LR scheduling.
- Achieved superior robustness scores for ConvNeXt-tiny.

Real-Time Face Mask Detection (2022-06-01 - 2022-08-01)https://github.com/SurajBhar/Face_Mask_Detection

- Achieved 93.5% accuracy in real-time prediction pipeline.
- Applied transfer learning on VGG16 features.

Languages

- English, Proficient (IELTS C1)
- German, Intermediate (B1 in progress)
- Hindi, Native

Interests**AI Research & Machine Learning**[Self-Supervised Learning](#) [Large Language Models](#) [Generative Models](#) [Transformers](#) [Deep Learning](#)**Computer Vision & 3D Perception**[Object Detection](#) [NeRF & DIVER](#) [Voxel-based Models](#) [LiDAR Systems](#) [Point Cloud Processing](#)**MLOps & Scalable Systems**[Model Deployment](#) [CI/CD](#) [Docker](#) [AWS](#) [MLflow](#)

References**David Lerch**

Research Scientist, Perceptual User Interface Group, Fraunhofer IOSB, Karlsruhe, Germany. Email: david.lerch@iosb.fraunhofer.de

Prof. Michael Moeller

Head of Computer Vision Department, University of Siegen. Website: <https://sites.google.com/site/michaelmoellermath/>

Prof. Dr. Roman Obermaisser

Full Professor, Division for Embedded Systems, University of Siegen. Profile: <https://blogs.uni-siegen.de/ms-cps/prof-dr-roman-obermaisser/>